

Lecture 7

1. Expected Value: "center of ^{mass of} the distribution"

Example: toss a coin twice. $X = \#$ of heads

$$P(X) = \begin{cases} 1/4 & \text{if } x=0 \\ 1/2 & \text{if } x=1 \\ 1/4 & \text{if } x=2 \end{cases}$$

$$E[X] = 0 \cdot \frac{1}{4} + 1 \cdot \frac{1}{2} + 2 \cdot \frac{1}{4} = 1$$

DEF: Suppose X is a bounded r.v. ("The values X can take on are finite") assuming the values $x_1, x_2, x_3, \dots$ with corresponding p.m.f. values $p(x_1), p(x_2), p(x_3), \dots$. The mean or expected value of X is given by

$$E[X] = \sum_n x_n \cdot p(x_n)$$

Example: Suppose we roll a die. If it comes up even, lose \$4
If it comes up 1, win \$1
3, win \$3
5, win \$5

$E[X]$ if X is your earnings?

$$E[X] = 1 \cdot \frac{1}{6} + 3 \cdot \frac{1}{6} + 5 \cdot \frac{1}{6} - 4 \cdot \frac{1}{2} = -\frac{1}{2}$$

Example: Suppose you pick 3 different numbers from 1 to 12. If you pick all 3 correctly, you win \$100. What are your expected earnings if it costs \$1 to play?

$X =$ your earnings

Outcomes: winning or losing

Earnings assoc. w/ the outcomes: \$99 or -\$1

$$P(\text{winning}) = \frac{1}{\binom{12}{3}} = \frac{1}{220}$$

$$P(\text{losing}) = 1 - P(\text{winning}) = 1 - \frac{1}{220} = \frac{219}{220}$$

$$E[X] = 99 \cdot \frac{1}{220} - 1 \cdot \frac{219}{220} = -\frac{120}{220} = -0.55$$

DEF: (Rigorous definition of expected value of discrete r.v.s) Let X be a discrete r.v. with pmf p . If one of the following sums is finite:

$$\sum_{x: x \geq 0} x \cdot p(x) \quad \sum_{x: x < 0} x \cdot p(x)$$

Then we say the mean/expected value (expectation of X exists and is defined to be

$$\mathbb{E}[X] = \sum_x x \cdot p(x)$$

If both sums are infinite, then $\mathbb{E}[X]$ does not exist

e.g., let X be a discrete r.v. with p.m.f.

$$p(x) = \begin{cases} \frac{1}{2|x|(1+|x|)} & \text{if } x = \pm 1, \pm 2, \dots \\ 0 & \text{otherwise} \end{cases}$$

We have that $p(x)$ is a valid pmf } exercise
But $\mathbb{E}[X]$ does not exist

We show that $\sum_{x=-1}^{-\infty} x \cdot p(x) = -\infty$ $\leftarrow$ e.g. $\sum_{x=-1}^{-\infty} \frac{x}{2|x|(1+|x|)} \sim \sum_{x=-1}^{-\infty} \frac{1}{x} \rightarrow -\infty$

$$\sum_{x=1}^{\infty} x \cdot p(x) = \infty$$

DEF: (Expectation of continuous r.v.s) If X is a bounded r.v. with p.d.f. f
The expectation of X is given by

$$\mathbb{E}[X] = \int_{\mathbb{R}} x \cdot f(x) dx$$

e.g., Suppose X is a continuous r.v. with pdf

$$f(x) = \begin{cases} 2x & 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

$$\mathbb{E}[X] = \int_{\mathbb{R}} 2x \mathbb{1}_{\{0 < x < 1\}} dx = \int_0^1 2x dx = \frac{2}{3}$$

DEF: (More rigorous if X is unbounded, continuous) Suppose X is a continuous r.v. with pdf f . Suppose that one of the following integrals is finite

$$\int_0^{\infty} x \cdot f(x) dx \quad \int_{-\infty}^0 x \cdot f(x) dx$$

Then the mean/expected value (expectation of X exists and is given by

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x \cdot f(x) dx$$

If both integrals are infinite then $\mathbb{E}[X]$ does not exist

Expected value of a function of a random variable

Q: What is $\mathbb{E}[g(X)]$ for some g ?

Approach 1: Define $Y = g(X)$. Find distribution (pmf or pdf) of Y and then calculate $\mathbb{E}[Y]$ (verify that $\mathbb{E}[Y]$ exists)

Approach 2 (what we will do): Law of the Unconscious Statistician (LOTUS)

THM: If X is discrete with pmf p , then $\mathbb{E}[g(X)] = \sum_k g(x_k) p(x_k)$ *

If X is continuous with pdf f , then $\mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x) f(x) dx$ *

* Provided that $\mathbb{E}[g(X)]$ exists

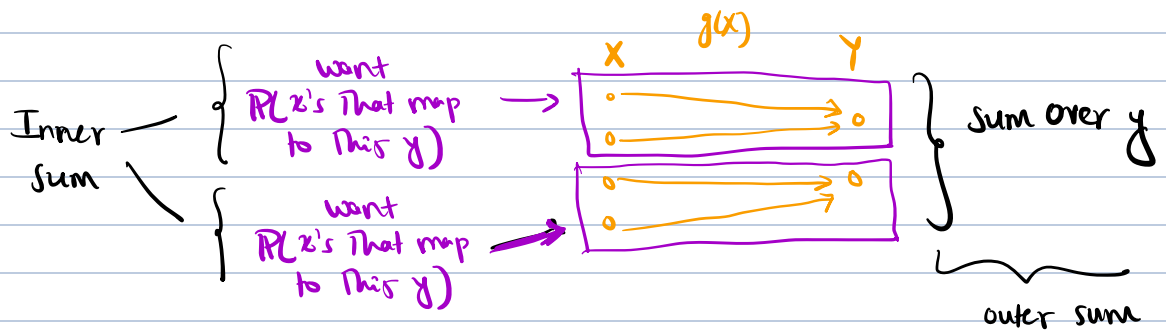
Proof: (Discrete case) Let $Y = g(X)$ and p_Y be the pmf of Y

$$\mathbb{E}[g(X)] = \mathbb{E}[Y] \stackrel{\text{DEF}}{=} \sum_y y p_Y(y)$$

$$\stackrel{\text{DEF}}{=} \sum_y y \cdot \mathbb{P}(Y=y)$$

$$= \sum_y y \mathbb{P}(g(X)=y) = \sum_y y \cdot \sum_{x: g(x)=y} \mathbb{P}(X=x)$$

$$= \sum_y \sum_{x: g(x)=y} g(x) \mathbb{P}(X=x) \stackrel{?}{=} \sum_x g(x) p(x)$$



Exercise: Roll a die. Let X be the # that comes up

- $\mathbb{E}[X^2]$
- $\mathbb{E}[\sqrt{X}]$
- $\mathbb{E}[e^X]$